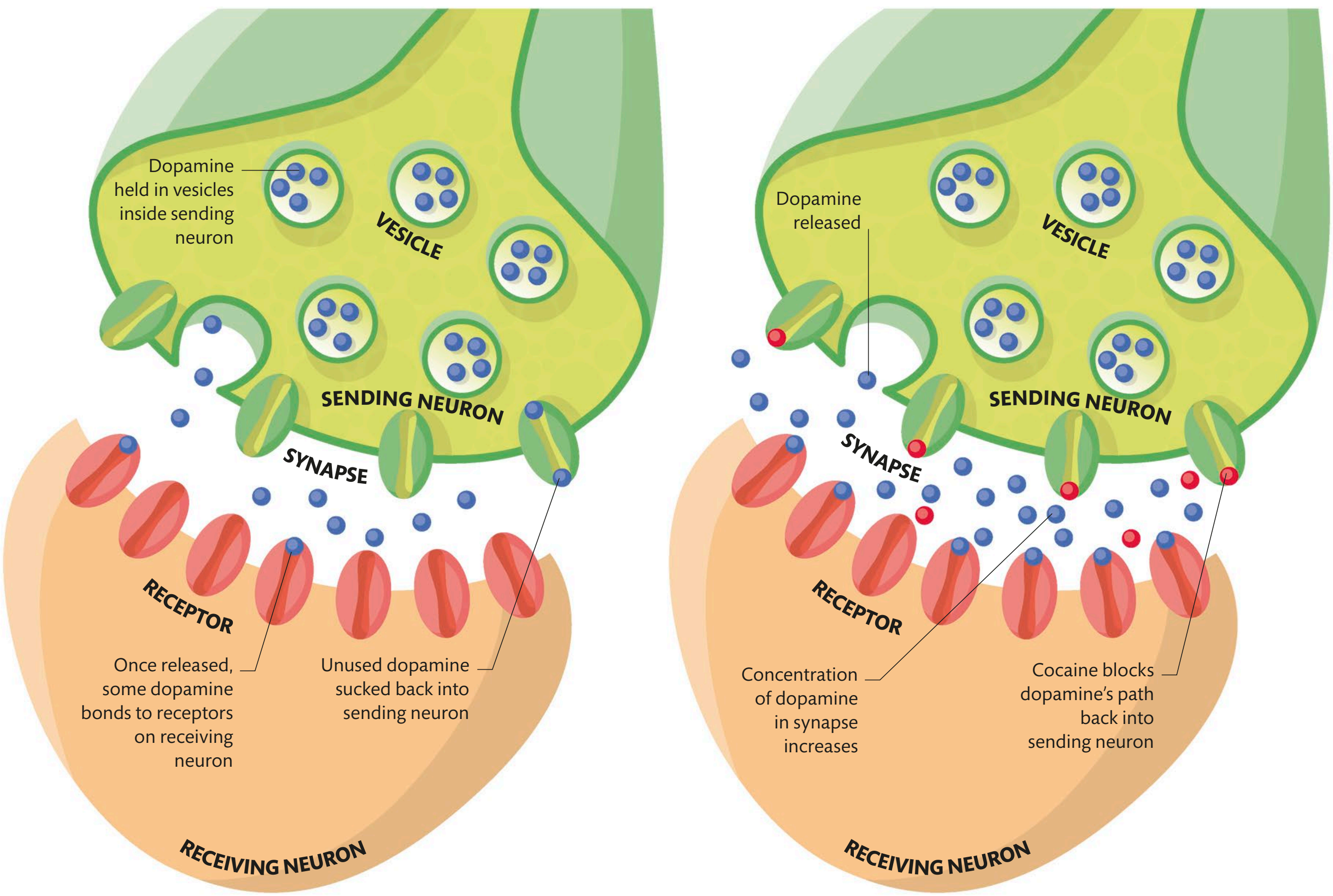
THE LONG-TERM EFFECTS OF ALCOHOL

Drinking large volumes of alcohol over a long period alters mood, arousal, behavior, and neuropsychological functioning. Alcohol's depressant effect both excites GABA and inhibits glutamate, decreasing brain activity. It also triggers the brain's reward centers by releasing dopamine, in some cases leading to addiction.

KEY
- Dopamine
- Cocaine

Dopamine and cocaine
The effects of cocaine are a product of its effects on the neurotransmitter dopamine at synapses in the brain.



Normal dopamine levels
Dopamine is a neurotransmitter associated with feeling pleasure. It creates a drive to repeat certain behaviors that trigger feelings of reward, perhaps leading to addiction. While some dopamine molecules bind to receptors on the receiving neuron, unused dopamine is recycled by being pumped back into the sending neuron and parceled up again.

With use of cocaine
Cocaine molecules are reuptake inhibitors of dopamine. When dopamine is released, it moves into the synapse and binds to receptors on the receiving neuron as normal. However, the cocaine has blocked the reuptake pumps that recycle the dopamine, so the neurotransmitter accumulates in a higher concentration, increasing its effects on the receiving neuron.